



يونيسف سلطان زين العابدين
unisza
UNIVERSITI SULTAN ZAINAL ABIDIN

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LECTURER : AHMAD FAISAL AMRI BIN ABIDIN @ BHARUN

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NAME	MATRICS NUMBER
ELYANA INSYIRAH BINTI MOHD SUHAIMI	076967

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1.0 INTRODUCTION



FIGURE 1: *Digital Forensics Investigation Process*

Digital forensics is a branch or an extension of forensic science which includes the process of identifying, preserving, analyzing, documenting and presenting digital evidence. It prioritizes the recovery and investigation of the data identified within the digital evidence that are related to the cybercrime as evidence to be presented later in court [1]. Nowadays, with organizations relying more and more on digital infrastructure, cybercrimes pose the biggest threats with the possibility of attacks compromising the organization's important digital assets, employees' and stakeholders' data as well as the organization's intellectual property. One of those threats include insider threats which points to threats from those who have legitimate access to the company's system, thus allowing them to pass through some of the security controls and can exfiltrate data with little to no detection [2].

When it comes to digital forensics, digital evidence preservation is highly crucial as its integrity and authentication must be proven flawless, else there is a chance the case would end up getting dismissed in court. According to the National Institute of Standards and Technology (NIST), forensic processes must ensure that digital evidence remains unaltered, well-documented and can be reproduced for maintaining its admissibility in legal proceedings [3]. This includes strict adherence to procedures such as chain of custody, hashing for integrity verification, and the use of validated forensic tools.

This laboratory report examines a simulated insider data breach scenario involving unauthorized access to confidential data, suspected data exfiltration via USB and cloud

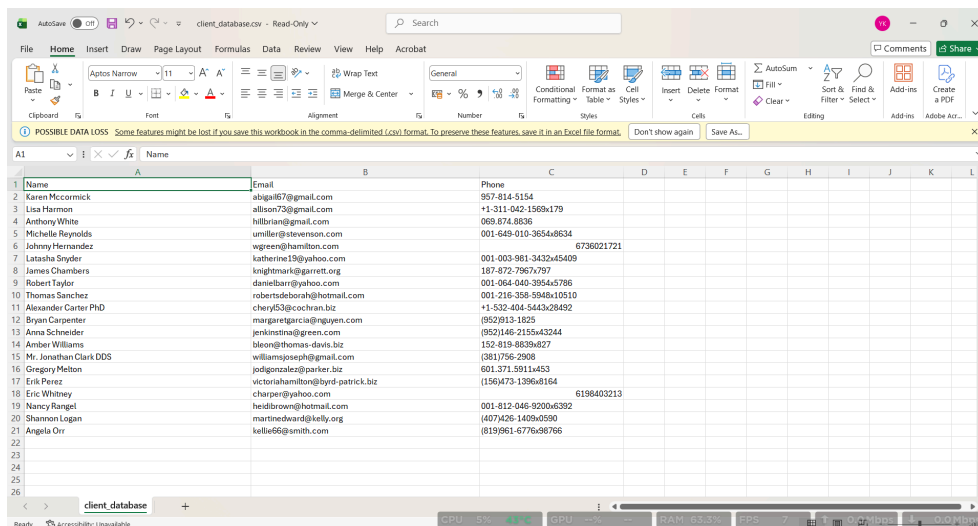
platforms and attempts to delete digital traces. The objective of this report is to apply professional forensic methodologies to prioritize evidence, respond effectively to a live incident, ensure legal compliance and maintain the integrity of digital evidence throughout the entire investigation process.

2.0 SCENARIO INTERPRETATION

The incident involves a suspected insider named Ahmad Fauzi, who allegedly accessed confidential company data and exfiltrated it using external cloud platforms (Dropbox and Google Drive) and a USB device.

Based on the findings, the incident likely occurred on 12th March 2025. The timeline is as shown:

- **13:00:** Suspect (Ahmad Fauzi) logged into the workstation.
- **14:00:** Suspect accessed 2 sensitive documents. `client_database.csv` which contains Clients' name, email and phone number and `salary_records.csv` which contains clients' name and their salary.



	A	B	C	D	E	F	G	H	I	J	K	L
	Name	Email	Phone									
1	Name	Email	Phone									
2	Karen McCormick	abigail67@gmail.com	957-814-5154									
3	Lisa Harmon	alison73@gmail.com	+1-311-042-1569x179									
4	Anthony White	hillbrian@gmail.com	065.874.8836									
5	Michelle Reynolds	umiller@stevenson.com	001-649-010-365x48634									
6	Johnny Hernandez	wgreen@hamilton.com	6736021721									
7	Latasha Snyder	katherine19@yahoo.com	001-003-981-3432x45409									
8	James Chambers	knightrick@garrett.org	187-872-7967x797									
9	Robert Taylor	danielbarr@yahoo.com	001-064-040-395x5786									
10	Thomas Sanchez	robertsdeborah@hotmail.com	001-216-358-5948x10510									
11	Alexander Carter PhD	chery63@cochran.biz	+1-532-404-5443x28492									
12	Bryan Carpenter	margaregarcia@greene.com	(952)915-1825									
13	Anna Schneider	jenkins@green.com	(952)146-2155x43244									
14	Amber Williams	bleon@thomas-davis.biz	152-819-8839x827									
15	Mr. Jonathan Clark DDS	williamsjoseph@gmail.com	(381)756-2908									
16	Gregory Melton	jodigonzalez@parier.biz	601.571.5911x453									
17	Eric Perez	victoriahamilton@byrd-patrick.biz	(156)473-1396x164									
18	Eric Whitney	charper@yahoo.com	6198403213									
19	Nancy Rangel	heidbrow@hotmail.com	001-812-046-9200x6392									
20	Shannon Logan	martinedward@kelly.org	(407)426-1469x4590									
21	Angela Orr	kellie6@gmail.com	(819)861-6776x96766									
22												
23												
24												
25												
26												

FIGURE 2: `client_database.csv`

Name	Salary
Audrey Newton	8888
Melissa Guerra	12239
William Evans	8034
Jeweller Wilson	13347
Jamie Blankenship	6360
Casey Freeman DDS	14139
Melissa Ruiz	12711
Travis Wallace	14173
Jessica Fleming	3194
Brett Andrade	9830
Eric Fisher	12043
Amanda Fields	12788
Dana Brown	6165
Nathan Gonzales	3830
Brent Montgomery	3680
Norma Lambert	9779
William Crawford	4818
Heather Saunders	4002
Donald Smith	5867
Brandon Morgan	7546

FIGURE 3: *salary_records.csv*

- **14:32:** A 32GB SanDisk USB was inserted.
- **Ongoing:** Suspect attempted to upload files via Dropbox and Google Drive, sent a suspicious email regarding an "attached file" and attempted to wipe CSV data (*client_database_deleted.csv*).
- Existence of a .pcap (packet capture) file suggests that the system is still powered on with active network communication.

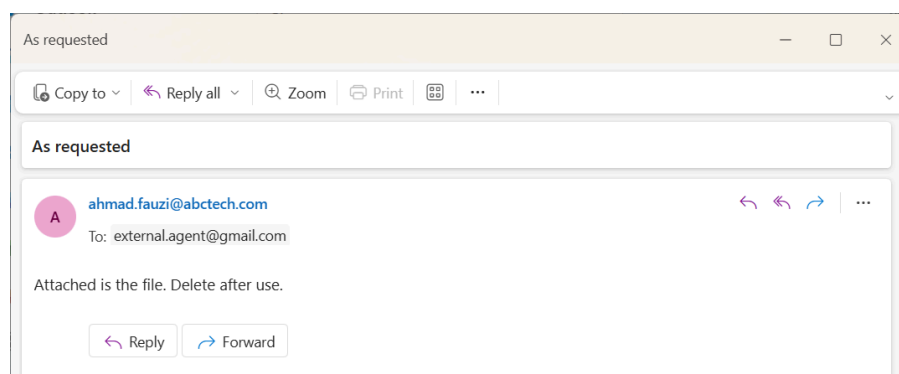


FIGURE 4: *Suspicious email of suspect with unknown personnel*

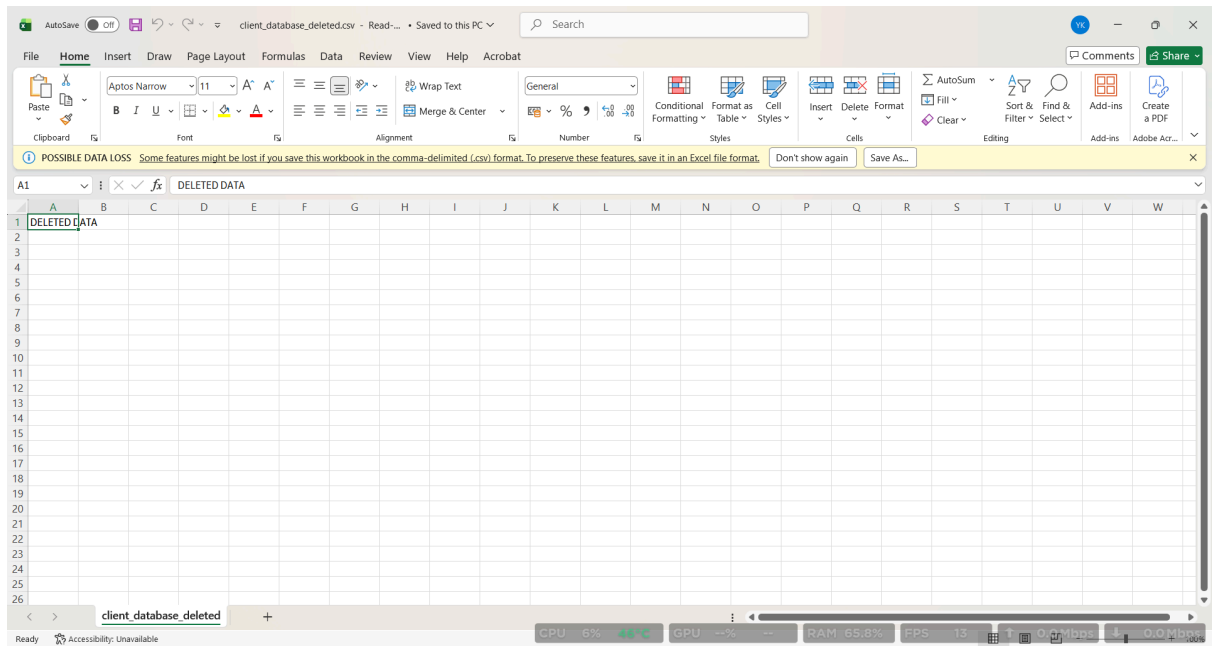


FIGURE 5: *client_database_deleted.csv*

These evidences highly indicate data exfiltration and attempt to cover the trace by deleting the .csv file.

3.0 PROFESSIONAL ANALYSIS

PART A: Evidence Prioritization.

3.A.1 Identification of Potential Sources of Digital Evidence

Multiple sources of digital evidence were found and identified, then categorized based on their volatility and storage type.

- **Volatile (Live) Evidence**

- **Random Access Memory (RAM)**

- Contains running processes, encryption keys, clipboard data and possible traces of data exfiltration.

- **Active Network Connections**

- Includes ongoing communications with external servers (cloud platforms or email services).

- **Current Logged-in Session**

- User activity, open applications and authentication tokens.

- **Connected USB Device (SanDisk 32GB)**

- May contain recently transferred confidential data and residual metadata.

- **Non-Volatile (Static) Evidence**

- **Hard Disk Drive (HDD/SSD)**

- Client database (client_database.csv).

- Salary data (salary_records.csv).

- Deleted file remnants (client_database_deleted.csv).

- **Browser History**

- Visits to cloud storage platforms (Dropbox, Google Drive).

- **Email Artifacts**

- Suspicious email indicating possible data transfer intent.

- **System Logs**

Login records and file access timestamps.

3.A.2 Evidence Prioritization Based on Volatility

Following the **order of volatility principle** [7], evidence is prioritized as follows:

Priority	Evidence Type
1	RAM
2	Active Network Connections
3	USB Device (connected)
4	Live System State (user session)
5	Disk-Based Evidence (files, logs, emails)

TABLE 1: *Evidence Prioritization Based on Volatility (Highest [1] to Lowest [5])*

3.A.3 Professional Justification

(a) RAM – Highest Priority

Volatile memory is the most critical yet unstable source of evidence. This is due to it containing transient data that, when the PC is switched off or interrupted, all previously stored information will disappear or gone [5] .Those information includes:

- Running processes
- Network sessions
- Encryption keys
- Fileless malware

Memory forensics is important because it reveals evidence that is not available on disk, such as live system activity and hidden threats. However, RAM data is also highly unstable and can disappear instantly, making immediate acquisition during live investigations very necessary [5].

A 2022 study by Kumar, R., & Tripathi, R. also emphasizes that live RAM data becomes the most important evidence when a system is running, as it directly reflects ongoing criminal activity [6].

(b) Network Activity – Second Priority

Live network traffic may contain:

- Evidence of ongoing data exfiltration
- Communication with external servers
- Upload attempts (cloud platforms)

Studies from [5] show that volatile memory and live system analysis can expose active internet connections and real time attacker behavior, which are otherwise unrecoverable after shutdown .

(c) USB Device – Third Priority

The connected USB device acts as a connector or bridge between internal data and external exfiltration with the possibility of containing these items below:

- Copied confidential files
- Timestamps matching system logs
- Residual traces of file transfer

Volatile evidence research shows that removable media often plays a direct role in data exfiltration and insider threats, thus making it the most suspected physical artifact to be closely tied to the incident timeline [7].

(d) Active System State – Fourth Priority

The currently logged-in session provides contextual evidence such as:

- User identity (Ahmad Fauzi)
- Opened files and applications
- Active browser sessions

Memory forensics studies confirm that analyzing live systems provides a snapshot of system state, including user activity and running applications at a specific time [7] .

(e) Persistent Storage – Lowest Priority

Disk-based evidence includes:

- CSV files (client database, salary data)
- Deleted file remnants
- Browser and email records
- System logs

Although these files are important, they are non-volatile and would remain intact even after the machine got shut down or rebooted. Research in [5] confirms that disk data is stable compared to volatile memory, making it less sensitive to time during initial response .

3.A.4 Key Forensic Principle Applied

The prioritization follows the Order of Volatility, a core forensic concept stating that:

“Evidence that is most likely to change or disappear must be collected first.”

This principle is strongly supported by modern digital forensics research [5], which emphasizes rapid acquisition of volatile data to avoid irreversible evidence loss .

PART B: Real-Time Crime Scene Response Strategy

3.B.1 Real-Time Response

Upon arrival at the crime scene, I'll prioritize establishing control over the environment and prevent any loss or contamination of evidence.

There, it is observed that a workstation is powered on with a USB device connected to it as well as an ongoing network activity, which strongly suggests potential live data exfiltration. Multiple employees are present, increasing the risk of accidental or intentional interference.

I immediately instruct all personnel to:

- Step away from nearby systems
- Cease any interaction with digital devices
- Remain available for questioning

This is necessary to preserve the current state of digital evidence and prevent contamination.

3.B.2 Critical Decision: Live Acquisition vs System Shutdown

Decision: Because the browser history shows active uploads to Dropbox/Drive, perform Live Acquisition.

3.B.3 Professional Justification

I decided to perform live acquisition instead of system shutdown due to shutting down the system at this stage would result in the irreversible loss of volatile evidence, including:

- Contents of RAM
- Active processes
- Network connections

- Possible decryption keys

Live digital forensic acquisition is noted to be very important when systems are running, as volatile data can provide critical insight into ongoing attacks and cannot be recovered after shutdown. RAM analysis can also reveal hidden processes, attacker activity and live network sessions, making it a primary evidence source in active incidents [5][7].

3.B.4 Real-Time Action Execution

After securing the scene, I will proceed with a controlled and minimal interaction with the workstation:

1. Scene Documentation

- Photograph workstation, USB device and screen display
- Record system time and visible activity
- Note physical connections (USB, network cables)

This will ensure that the original state of the scene is preserved for legal and analytical purposes.

2. Network Isolation (Without Power Loss)

Subsequently, I will isolate the system from the network by:

- Disconnecting Ethernet cable or disabling Wi-Fi

This step is critical to stop ongoing data exfiltration, prevent remote tampering and preserve internal system state. Research indicates that isolating a system while keeping it powered on allows investigators to contain threats without losing volatile evidence [7].

3. Evidence Preservation Controls

To prevent contamination:

- I avoid using the system unnecessarily
- I ensure only forensically sound tools are used
- I document every action performed

Any improper handling at this stage could alter timestamps and overwrite memory making the evidence lose its integrity thus compromising legal admissibility.

4. Maintaining Forensic Integrity

To maintain forensics integrity, I will do:

- Chain of custody documentation
- Unique identification of each evidence item
- Logging of all interactions with the system

Through maintaining forensic integrity, it is ensured that evidence remains credible, reproducible and admissible in court. Modern forensic guidelines emphasize that improper handling or undocumented actions can lead to evidence being rejected in legal proceedings.

These proposed approach is effective because:

- Preserves high-value volatile evidence
- Prevents ongoing data exfiltration
- Minimizes risk of contamination
- Maintains legal admissibility

Thus, the decision to perform live acquisition reflects a professional forensic judgment, prioritizing the preservation of volatile evidence while maintaining control over the environment. By isolating the system without shutting it down, the investigation ensures that critical real time data is retained, enabling a more accurate reconstruction of the suspected data breach.

PART C: Adaptive Evidence Collection Plan

3.C.1 What to Collect FIRST and WHY

FIRST: RAM (Volatile Memory Acquisition)

The first thing to collect is RAM from the live system because the evidence contained here is the most unrecoverable and can easily disappear within time. This includes:

- Running processes (file transfer tools)
- Active network connections
- Decryption keys
- Evidence of file access not written to disk

Based on research from [5], it is confirmed that RAM contains “network connections, encryption keys, running processes and open files” that are often not recoverable from disk. Not to mention that memory forensics can still provide evidence even when other data has been deleted or wiped.

3.C.2 Adaptive Evidence Collection Strategy

(a) Live System (Running Workstation)

For live system or running workstation, my approaches are to:

- Perform RAM dump immediately
- Capture running processes and system state
- Record active user session

These are done because live system analysis provides a snapshot of system activity at that exact moment, which is needed and required to reconstruct an attacker's behavior. Research in [8] shows that memory forensics enables investigators to identify hidden or fileless

malware and recover execution traces not stored on disk. This is especially relevant because the suspect attempted to delete evidence while the system is still active.

(b) USB Device (SanDisk 32GB)

As for USB device, I will be:

- Physically secure the USB after documentation
- Use a write blocker before acquisition
- Create a forensic image of the device

This matters since the USB device is highly relevant. As for why, it is due to the fact that it was inserted shortly after accessing a sensitive file and has a likelihood of containing an exfiltrated data. Although USB evidence is non-volatile, it is still at high risk of tampering or removal in a live scene. Memory forensics research also highlights that external devices and their activity can leave traces in RAM, thus proving the importance of correlating RAM + USB evidence together [5].

(c) Network Traffic (Ongoing Communication)

When it comes to dealing with network traffic or ongoing communication, I will have to:

- Capture network traffic if possible (packet capture)
- Record active connections before isolation
- Identify connections to:
 - Cloud services (Dropbox, Google Drive)
 - External email recipient

This is due to RAM analysis often reveals live network sessions and remote connections that may not be stored persistently [5]. It is to add that cloud uploads may leave minimal disk traces, making network evidence the only proof of exfiltration.

3.C.3 Forensic Tools Selection & Justification

Tool 1: FTK Imager

Its purpose is for Disk and USB imaging. FTK Imager is suitable for this case as it creates forensically sound bit-by-bit copies which is the type of evidence that is widely accepted in courts. However, it does not come without risk. If it is misused, as in doing imaging without a write blocker, can alter the evidence. Improper configuration can also lead to incomplete acquisition.

Tool 2: Volatility

Volatility is used for RAM analysis. It extracts running processes, network connections and command execution traces. Research has shown that memory forensics tools can reconstruct system activity and attacker behavior from RAM artifacts [5]. It is to note that if misused, like providing incorrect memory profile would result in inaccurate result. Misinterpretation can also lead to false conclusions, making the evidence loses its legal admissibility.

Tool 3: Wireshark

Wireshark is widely known and used for network traffic capture. It can help detect data exfiltration through any suspicious outbound connections that were captured. However, late packet capture can lead to missing evidence, making timing highly crucial. Improper filtering also poses a risk of data overload or missed indicators.

3.C.4 Professional Decision-Making & Adaptability

This plan demonstrates adaptability because it:

- Adjusts based on live vs static evidence
- Prioritizes high risk evidence first

- Integrates multiple evidence sources (RAM + USB + Network)

Modern cyber investigations require correlation across volatile memory, external devices and network activity, as attackers increasingly use techniques that leave minimal traces on disk, causing memory captures to seem to be hidden and transient activity [5]. Not to mention that disk alone is also insufficient for modern investigations [8].

The evidence collection strategy prioritizes volatile data while adapting to the presence of external storage and live network activity. By integrating RAM acquisition, USB imaging and network monitoring, the approach ensures comprehensive evidence recovery while minimizing the risk of data loss, contamination and incomplete forensic reconstruction.

PART D: Legal Risk Assessment & Compliance

3.D.1 Legal Readiness of the Investigation

Before conducting any forensic investigation, I must ensure that the investigation is legally authorized, properly scoped and compliant with privacy obligations.

(a) Authorization

In this scenario, the investigation involves:

- Employee data (client database, salary records)
- Personal communications (email)
- Potential external data transfer

For this reason, I must obtain formal authorization from the organization like management or legal department and if required, law enforcement approval or warrant (if external parties are involved).

(b) Scope of Investigation

The investigation scope is limited strictly to:

- Activities related to Ahmad Fauzi
- Files involved in the suspected breach (CSV files, email, USB activity)

No access to unrelated employee data, even if it is available on the system, should be done as it is not covered within the scope.

(c) Privacy Considerations

The dataset related to the incident are:

- Client personal information (names, emails, phone numbers)

- Employee salary data

Under Malaysia's Personal Data Protection Act 2010, these information are categorized as personal data and improper handling can result in legal consequences [9]. For this reason, I must:

- Limit access to only necessary data
- Avoid unnecessary duplication or exposure
- Ensure secure handling of all personal information

3.D.2 Legal Risks if Procedures Are Not Followed

Risk 1: Evidence Becomes Inadmissible in Court

If proper forensic procedures are not followed due to things like not providing documentation and including altered evidence, the defense has the right to challenge the integrity of evidence. If successful, the case may be dismissed [10]. For example in this specific case, if the USB device is accessed without proper handling, it may be argued that the evidence was modified after seizure.

Risk 2: Violation of Privacy Laws

Accessing employee emails as well as personal client data without proper authorization may violate the Personal Data Protection Act 2010, leading to legal penalties and organizational liability [9]. For example reading unrelated emails beyond the investigation scope could be considered unauthorized data access.

Risk 3: Evidence Contamination or Tampering

Improper handling like no usage of write blocker or undocumented actions have the possibility of altering timestamps, files modification and original evidence deletion. This compromises both evidence reliability and investigator credibility.

Risk 4: Exceeding Investigation Scope

Investigating beyond Ahmad Fauzi's actions like checking other employees without cause may violate internal policies and can lead to legal challenges.

3.D.3 Professional Mitigation Strategies

(a) Ensure Proper Authorization

To ensure investigation is done with proper authorization, I will need to obtain written approval before investigation and ensure I clearly define what systems can be accessed and what data can be analyzed. Otherwise, my intention and eligibility might be questioned.

(b) Strict Documentation

Digital forensics investigation requires every action taken to be well-recorded and documented. This is to ensure time logs and evidence handling records are properly maintained, thus supporting transparency and legal defensibility.

(c) Apply Forensic Best Practices

Some of the digital forensic best practices is to use write blockers as it allows data access and permission to read from a storage device without altering it. Others include performing forensic imaging instead of direct access to avoid altering original evidence.

(d) Data Minimization Principle

Data Minimization Principle refers to

“Only collect, access or process the minimum amount of data necessary to achieve a specific purpose.”

This means only relevant data can be collected to ensure compliance with privacy laws [9].

In this one specific case, it refers to:

- CSV files
- USB activity
- Email communication

(e) Secure Evidence Handling

To secure evidence handling as to avoid any tampering, evidence must be stored in controlled environments and access has to be restricted to authorized personnel only. This is to avoid any form of accidental or non-accidental alteration to the evidence.

Digital forensic investigations are both technical and legal processes. This means that even if evidence proves guilt, it is useless in court if obtained improperly and investigators themselves may face legal consequences. The legal success of this investigation depends not only on identifying digital evidence but also on ensuring that all actions are properly authorized, documented and compliant with privacy laws. By maintaining strict adherence to legal and forensic standards, the investigation preserves both the integrity of the evidence and its admissibility in legal proceedings.

PART E: Evidence Integrity & Preservation Strategy

3.E.1 Chain of Custody Process

The chain of custody ensures that digital evidence is tracked, controlled and documented in chronological order from the moment it is collected until it is presented in court.

Process Implementation

For this case, each item (workstation, SanDisk USB, forensic images) will be:

- Assigned a unique evidence ID
- Labeled with:
 - Date and time of collection
 - Location
 - Investigator's name
- Logged in a chain of custody form, recording:
 - Every transfer
 - Every handler
 - Purpose of access

According to the National Institute of Standards and Technology, maintaining proper documentation of evidence handling is essential to ensure its authenticity and admissibility in legal proceedings [10].

3.E.2 Evidence Documentation Standard

All actions taken during the investigation must be fully documented to ensure transparency and repeatability.

Documentation Includes

- Scene photographs (workstation, USB, screen state)

- Notes on system status (powered on, active session)
- Time synchronization (system time vs actual time)
- Detailed activity logs (every forensic action performed)

NIST guidelines [10] emphasized that documentation must be very detailed to allow another examiner to reproduce the results independently.

3.E.3 Hashing Strategy (MD5 & SHA-256)

Hashing is used to verify that evidence remains unchanged throughout the investigation. The implementation of hashing is as below:

- Generate hash values:
 - Before acquisition (if possible)
 - After imaging
 - Before and after analysis
- Use:
 - MD5 for quick integrity check
 - SHA-256 for stronger and more secure verification

NIST states that cryptographic hash functions are essential for ensuring data integrity and detecting any modification of evidence [11].

3.E.4 Secure Storage Mechanism

After acquisition, evidence must be stored securely to prevent unauthorized access, physical damage and tampering that can alter or damage the evidence. Below are strategies to store evidence securely:

- Store original devices in:
 - Tamper-evident bags

- Locked forensic evidence cabinets
- Store forensic images:
 - In secure servers with restricted access
 - With access logs enabled
- Maintain backup copies for redundancy

3.E.5 What Could Go Wrong if Integrity is Compromised

Failure or inability to preserve evidence properly can lead to several consequences like:

(a) Evidence Tampering Claims

Defense may argue that data was altered, resulting in the evidence becoming unreliable.

(b) Loss of Admissibility

The court may reject evidence entirely thus the case may end up getting dismissed.

(c) Investigation Credibility Damage

The investigator's professionalism will be questioned, leading to all of the findings being challenged and getting objected.

3.E.6 How This Strategy Prevents Issues

These approaches prevent issues from occurring because:

- Chain of custody will prove who handled evidence at all times
- Documentation ensures transparency and repeatability
- Hashing guarantees data integrity
- Secure storage prevents unauthorized modification

According to NIST forensic guidelines [10][11], combining these practices ensures that digital evidence remains authentic, reliable and legally defensible throughout the investigation process.

In digital forensics, integrity must not be assumed, but proven. Valid evidence becomes useless if it cannot be verified and is proven to be handled questionably. A forensically sound preservation strategy requires strict adherence to chain of custody procedures, comprehensive documentation, cryptographic hashing and secure storage practices. By implementing these controls, the investigation ensures that all digital evidence remains intact, verifiable and admissible in court, thereby maintaining both technical and legal integrity.

PART F: Professional Case Resolution & Improvement Plan (20 Marks)

3.F.1 Critical Evaluation of the Investigation Scenario

This incident is not just an individual insider breach, it reflects systemic organizational flaws and weaknesses in security controls, monitoring and policy enforcement. From the dataset, the breach followed a clear path:

Authorized access → Data extraction → External transfer (USB/cloud) → Attempted concealment

While the employee (Ahmad Fauzi) is responsible for the act, the organization failed to:

- Restrict unauthorized data movement
- Detect abnormal user behavior
- Prevent data exfiltration channels

This indicates a lack of defense-in-depth strategy, where multiple layers of security controls should exist to prevent or detect such actions. According to the National Institute of Standards and Technology, organizations must implement layered security controls to reduce the risk of insider threats and unauthorized data access (NIST SP 800-53 Rev. 5) [12].

3.F.2 Identified Weaknesses (Root Cause Analysis)

Weakness 1: Unrestricted Use of Removable Media

A standard employee was able to:

- Insert a **SanDisk 32GB USB**
- Potentially copy sensitive CSV files

This indicates:

- No **USB port control policy**
- No enforcement of **device authorization**

Weakness 2: Lack of Data Loss Prevention (DLP) Controls

Sensitive data (client database, salary records) was accessed and potentially transferred without triggering alerts or restrictions.

This shows absence of:

- Content monitoring
- Data exfiltration prevention mechanisms

Weakness 3: No Cloud Upload Restrictions

Browser history shows access to:

- Dropbox
- Google Drive

This indicates:

- No firewall or proxy rules blocking unauthorized uploads
- No monitoring of outbound data traffic

Weakness 4: Weak Email Security Controls

The suspicious email:

“Attached is the file. Delete after use.”

suggests:

- No data leakage detection in email systems
- No monitoring of suspicious outbound communication

Weakness 5: Lack of User Activity Monitoring

Despite:

- File access at 14:00
- USB insertion at 14:32

No alerts were triggered.

This indicates absence of:

- User Behavior Analytics (UBA)
- Real-time monitoring systems

3.F.3 Potential Investigator Mistakes

Mistake 1: Immediate System Shutdown

Shutting down the system would:

- Destroy RAM evidence
- Terminate active network connections
- Remove traces of ongoing exfiltration

This contradicts forensic best practices, which emphasize preserving volatile data in live systems.

Mistake 2: Failure to Isolate the Scene

Allowing employees to interact with workstations may:

- Alter evidence
- Introduce new data
- Compromise chain of custody

Mistake 3: Accessing Devices Without Write Blocker

Directly opening the USB or disk may:

- Modify timestamps
- Create system files
- Invalidate evidence integrity

Mistake 4: Over-collection of Data

Collecting unrelated files like other employees' data is considered violating privacy principles and legal scope. This increases risk under data protection laws.

3.F.4 Weaknesses in Standard Procedures

(a) Over-Reliance on Static Forensics

Traditional approaches focus heavily on disk analysis, but this case involves live activity and deletion attempts. Thus, disk evidence alone may be incomplete

Research in [5] shows that volatile memory analysis is critical for detecting modern attack techniques, including fileless activity and transient processes.

(b) Lack of Real-Time Response Capability

Many procedures assume that when systems are powered off, evidence would remain static. This is unrealistic in modern environments where attacks occur in real time and data can be exfiltrated instantly.

(c) Weak Integration Between Evidence Sources

Failure to correlate:

- RAM

- USB
- Network
- Logs

can lead to incomplete reconstruction of events.

3.F.5 Improved Professional Investigation Workflow

Step 1: Secure and Control the Scene

- Remove unauthorized personnel
- Restrict access to devices
- Document environment

Step 2: Preserve Volatile Evidence

- Capture RAM immediately
- Record active processes and network connections

Step 3: Isolate System (Without Shutdown)

- Disconnect from network
- Prevent further exfiltration

Step 4: Seize and Preserve Physical Evidence

- Secure USB device
- Label and package properly

Step 5: Forensic Imaging

- Create bit-by-bit images of:
 - Disk

- USB device
- Verify using hash values

Step 6: Multi-Source Analysis

Correlate:

- System logs (file access at 14:00)
- USB logs (14:32 insertion)
- Browser history (cloud uploads)
- Email communication (intent)

Step 7: Timeline Reconstruction

Build a timeline to establish:

- Sequence of actions
- Intent
- Data movement

Step 8: Reporting & Presentation

- Document findings clearly
- Maintain legal defensibility
- Provide evidence-backed conclusions

3.F.6 Risk Mitigation Strategies

(a) Standardized Forensic Frameworks

Adopt structured methodologies such as:

- NIST forensic guidelines

- ISO/IEC 27037

These ensure consistency, reliability and legal compliance, making the evidence more likely to be accepted

(b) Continuous Investigator Training

Investigators must be trained in:

- Live forensics
- Memory analysis
- Cloud-related evidence

(c) Use of Validated Tools

Only use tools that are:

- Tested
- Widely accepted
- Forensically sound

(d) Strict Documentation Practices

Every action must be:

- Recorded
- Justified
- Reproducible

3.F.7 Justification of the Improved Approach

This workflow is more effective because it:

- Prioritizes volatile evidence preservation

- Prevents evidence contamination
- Integrates multiple evidence sources
- Ensures legal admissibility

And the improvements suggested align with modern cybersecurity principles such as:

- Defense-in-depth for multiple layers of protection
- Zero trust mindset to ensure no implicit trust in users
- Data-centric security to protect the data itself and not just the system

According to the National Institute of Standards and Technology, implementing layered security controls significantly reduces the risk of insider threats and data breaches (NIST SP 800-53 Rev. 5) [12].

Research and forensic standards emphasize that effective investigations require a combination of technical accuracy, procedural rigor and legal awareness. Since modern digital investigations are no longer linear and getting increasingly complex, they would require an adaptive, multi-layered analysis that combines live forensics, device analysis and network intelligence.

The effectiveness of a digital forensic investigation depends not only on technical tools but on the investigator's ability to make informed, real-time decisions while maintaining legal and procedural integrity. By addressing weaknesses in traditional approaches and implementing a structured, adaptive workflow, the investigation can accurately reconstruct the insider breach while ensuring that all evidence remains reliable, complete and admissible in court.

4.0 CRITICAL DISCUSSION

Investigating insider threats is more complex than dealing with external cyberattacks since the attacker operates within legitimate access boundaries. In this particular case, Ahmad Fauzi, who had authorized access to sensitive files, makes malicious activity harder to detect in real time and more difficult to distinguish from normal job functions.

One of the main challenges in this investigation is to balance between rapid incident containment and preservation of forensic integrity. On one hand, there is an urgent need to stop potential data exfiltration, especially since browser history indicates possible uploads to Dropbox and Google Drive. This is critical because delayed response can result in insensitive client and salary data to be permanently transferred outside the organization. On the other hand, forcefully shutting down the system or interacting excessively with live processes may alter volatile evidence such as RAM contents, active network sessions and running processes.

This creates a critical trade-off. Speed can protect the organization, but may destroy the important evidence while caution preserves evidence, but risks ongoing data loss. According to the National Institute of Standards and Technology [10], incident responders must try to balance containment actions with evidence preservation, especially in live systems where volatile data is present (NIST SP 800-86). Additionally, an immediate isolation of the system from the network is also recommended to prevent further compromise, while still maintaining the system in a powered on state for forensic acquisition.

Another challenge is ensuring forensic integrity under time pressure. In real world scenarios, investigators must make rapid decisions such as whether to:

- Capture RAM immediately
- Disconnect network access, or

- Secure removable media

Each action poses their own risk of evidence alteration if performed incorrectly. For example, a sudden unplugging of the system may stop data exfiltration but will destroy valuable volatile evidence needed to reconstruct attacker behavior.

Furthermore, insider investigations require heightened attention to scope control and privacy considerations. Since the investigator is dealing with legitimate user accounts and sensitive organizational data, excessive,uncontrolled and out of the scope data collection can lead to privacy violations and legal complications.

To conclude this section, insider threat investigations require a balance between operational urgency and forensic discipline. The investigator must act quickly enough to prevent data loss while ensuring that every action taken preserves the admissibility and integrity of digital evidence.

5.0 CONCLUSION

In conclusion, the investigation confirms that the most appropriate and reasonable response in this scenario is to perform live acquisition instead of immediate system shutdown. Since the workstation was powered on, actively used and showed evidence of possible ongoing network communication (including Dropbox and Google Drive access), shutting the system down would have resulted in the irreversible loss of volatile evidence such as RAM contents, active sessions and live network connections. These elements are critical in reconstructing the sequence of events and confirming whether cloud-based data exfiltration occurred.

Therefore, live forensic acquisition was the only logical approach to preserve both volatile system evidence and potential indicators of active data transfer, while still allowing investigators to contain the incident through controlled isolation measures. This ensures that the investigation captures the full context of the breach, including user activity, process execution and network behavior at the time of the incident.

At the same time, all handling of evidence containing personally identifiable information (PII), such as the client database and salary records, must strictly comply with legal and ethical requirements under the Personal Data Protection Act 2010 [9]. This includes ensuring that data is accessed only within the defined investigative scope, properly protected during storage and analysis and not exposed beyond what is necessary for forensic examination. Any sort of failure to comply with these requirements could result in legal liability and compromise the admissibility of evidence.

Overall, the case reinforces two key principles of digital forensics which are that live acquisition is essential in active breach scenarios involving cloud or network-based exfiltration, and that strict legal compliance must govern every stage of handling sensitive personal data to ensure both forensic integrity and regulatory adherence.

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